

SYMPOSIUM: EMERGING HEALTH ISSUES

Key Health Issues for Dairy Cattle—New and Old

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ABSTRACT

The objective of this paper is to use available information to evaluate the relative importance of various health issues affecting dairy cattle. In addition to traditional ranking using evaluation methods based on impacts to animal productivity, this paper considers zoonotic risks, international trade implications, and animal welfare concerns. Traditional production costs rank mastitis, reproductive problems, and lameness as the top dairy cattle diseases. When the other areas of importance are included, the top-ranked diseases change to include salmonella, Johne's disease, bovine viral diarrhea-associated disease, and mastitis. Researchers in the dairy industry may want to reevaluate their criteria for setting research priorities to include zoonotic risks, international trade implications, and animal welfare concerns.

(**Key words:** dairy cattle, animal health, disease)

Abbreviation key: BLV = bovine leukosis virus, BVD = bovine viral diarrhea, NAHMS = National Animal Health Monitoring System, OIE = Office International des Epizooties.

INTRODUCTION

Health issues affecting dairy cattle can be evaluated in many ways. The objective of this report is to use available information to evaluate importance of various health issues affecting dairy cattle, focusing on categories of impacts to animal productivity, zoonotic risks, international trade, and animal welfare. Health problems that have an impact on production returns through increased cost for treatment and reduced output may be classified as direct impacts because they can be observed by individual producers. Thus, producers can respond directly to the condition and also demand research activities and extension programs that will assist them in managing health problems that reduce productivity. Impacts of zoonotic risk, international trade, and welfare concerns, however, are more indirect and, thus, may be less

obvious to producers than their own direct costs. Producers may be less likely to demand action, even though the consequences may be greater than production losses because markets can be lost or gained.

For example, milk has a positive reputation of being a wholesome and health-promoting food, and most dairy operations are independent family farms where cows are well cared for. Combined, these two factors help form an image that is viewed very positively by consumers. However, if consumers perceive milk to be unsafe for zoonotic reasons or disapprove of animal husbandry practices, they can respond by consuming less dairy products. On the positive side, international markets offer the potential for increasing the number of people consuming US dairy products.

The importance of considering issues affecting both supply and demand is illustrated in a simple example. If improved animal health techniques lowered milk production cost by 5% and, thus, increased the milk supply, producers would benefit by an estimated \$534 million. Additionally, if the demand for milk increased by 5%, net returns of producers would rise by an estimated \$879 million. (These results were based on a demand-supply model reported in Ott et al. (26), using a demand elasticity of 0.61 (54), a supply elasticity of 0.642 (3), and an initial equilibrium farm price and quantity of \$14/cwt and 155,368 million pounds, respectively.) Thus factors affecting demand are at least as important to producers as factors affecting supply. For this reason, when evaluating the relative importance of various health issues affecting dairy cattle, we went beyond traditional impacts to animal productivity and included zoonotic risks, international trade, and animal welfare impacts as well.

Combining these different areas into a single, overall evaluation of the importance of specific dairy health conditions is subjective in nature and subject to debate. However, because of the limited resources available, research efforts should focus on the most important health conditions. Therefore, we attempted to prioritize dairy health conditions using a qualitative approach to combine the impacts from each of the four categories. To accomplish this prioritization, we used information from published literature when available; when not available, we used our best as-

Received June 23, 1997.

Accepted June 3, 1998.

assessments of available information for our analysis. For production losses, efforts were made to assess economic losses when data were available; when data were not available, conditions were ranked by estimated relative order. We then scored each health condition as 0, +, ++, or +++. For zoonotic risks, a similar process was used, with scores ranging from 0 to ++. For international trade and welfare concerns, health conditions were ranked either 0 or +. By allowing larger scores for production losses (+++) and zoonotic risks (++) , we were able to emphasize these impacts relative to the others in a manner consistent with our assessment of the importance of each category.

Health Conditions with High Impact on Production

The consequences of animal health on production cost have two components: 1) expenditures for veterinary services and drug supplies and 2) lost output. Dairy producers spent approximately \$580 million in 1993 on veterinary services and drug supplies (based on middle cost producers at \$60 per cow) (48). In addition to expenditures for treatment and prevention, diseases reduce the output of either milk or calves or require additional feed per unit of production.

Based on updates of previous studies to current prices and milk production levels (6, 12, 13, 16, 19, 20, 24, 25, 39, 52) and using information from the 1996 National Animal Health Monitoring System (NAHMS) Dairy '96 study (unpublished) we estimated the annual aggregated microcost (summation of individual producer costs) of mastitis in the US to be \$1.5 to 2.0 billion. Losses from subclinical mastitis of lost milk production and higher cow replacements costs associated with high somatic cell counts were estimated at \$960 million. Because of the recognition of the importance of mastitis, the vast majority of US dairy producers use recommended management practices for mastitis prevention such as teat dipping and dry cow treatment (42).

Reproductive diseases cost an estimated \$400 million, based on information collected nearly 10 yr ago but updated to current prices (39). Lameness is another important disease of dairy cattle; unfortunately, cost data for this disease are not readily available. These health conditions have been linked in varying degrees to reduced productivity, and preventive management strategies have been the focus of repeated research-based educational efforts.

Johne's disease, caused by *Mycobacterium paratuberculosis*, also has high costs. Based on

NAHMS Dairy '96 study (45), Johne's disease is estimated to cost US dairy producers almost 640 million kg of milk increased culling of 11,000 cows, and nearly 20,000 cow deaths annually (based on 1995 estimate of 9,458,000 dairy cows). In addition, of the estimated \$60 million in lost revenue for culls in poor body condition, \$9 million can be attributed to Johne's disease. Thus, the total amount of annual revenue lost from Johne's disease is approximately \$222 million.

From the NAHMS Dairy '96 study (45), the prevalence of US dairy herds infected with *M. paratuberculosis* was estimated to be at least 22%, although preventive strategies do exist. Johne's disease is usually introduced to dairy herds through the purchase of cattle that are infected but clinically normal (36). Thus, an important preventive strategy is to establish biosecurity measures that reduce the probability of infected cattle being introduced into the herd. The spread of disease within a herd can be reduced by preventing contamination of cattle feed or water by manure. On-farm preventive measures have been described (31). Unfortunately, many dairy producers are not familiar with Johne's disease. In the recent NAHMS study (45), only 55% of the producers surveyed stated that they knew some basics or were fairly knowledgeable about the disease. Thus, many herds are at risk from transmission between herds via introduction of infected cattle to the herd operation.

Another pathogen that could rank high in production costs is bovine viral diarrhea (BVD) virus. Economic losses from the BVD virus have not been estimated, and its varied spectrum of clinical presentations, including reproductive signs and other clinical signs, limits the recognition of BVD by producers. The prevalence of cattle exposure to this pathogen is considered to be high, partly because of widespread vaccination and partly because of viral transmission. Although information on the national prevalence in dairy cattle is unavailable, estimates from 256 beef cow-calf herds across the US (27) showed that 86% of unvaccinated herds and 95% of vaccinated herds were seropositive (at least one animal sampled within herd was antibody-positive). About 46% of beef cow-calf herds were reportedly unvaccinated. The BVD vaccines are used more commonly in dairy herds than in beef cattle herds (41), and, because of widespread exposure to BVD viruses (both vaccine and field strains), it is considered unlikely that large BVD epidemics of severe acute disease will occur in US dairy herds. Other clinical manifestations of BVD disease such as abortions and infertility are of concern, however. Use of approved BVD vaccines according to label is one method for

controlling disease associated with BVD. The BVD vaccination programs, however, are less effective when label directions are not followed. This reduced effectiveness is a concern because, although 71% of US dairy producers reported use of BVD vaccine in cows and 70% in replacement heifers, only 47% of the producers reported vaccinating in a manner that was consistent with label guidelines (41).

Health Conditions with Zoonotic Risk

A recent summary of costs of selected bacterial foodborne pathogens in humans in the US estimated *Salmonella* spp. costs at \$0.6 to 3.5 billion, *Campylobacter jejuni* or *Campylobacter coli* at \$0.6 to 1.0 billion, *Escherichia coli* O157:H7 at \$0.2 to 0.6 billion, and *Listeria monocytogenes* at \$0.2 to 0.3 billion (9). The total actual cost of annual bacterial foodborne disease in the US is \$2.9 to 6.7 billion. Estimates for cryptosporidiosis in humans are not available, but the potential for economic costs to society is high, as evidenced by the outbreak of cryptosporidiosis in Milwaukee, Wisconsin during spring 1993 that affected an estimated 403,000 persons (21).

Salmonella spp., *C. jejuni*, *E. coli* O157, *L. monocytogenes*, *Cryptosporidia parvum*, and *Giardia* spp. are human pathogens that are found in various types of foods and water sources, including foods of animal origin. All of these organisms, except *E. coli* O157, also are pathogenic to animals. One source of human exposure to these pathogens is beef and dairy cattle. Humans become infected primarily through fecal contamination of food products or water. Because of the widespread pasteurization of milk and dairy products, the risk to public health from these pasteurized products is small. Dairy cattle contribute about 25% of nonfed beef available for consumption in the US (33), however, and the risks from contaminated dairy beef products are a concern. Because of these risks, a key issue facing the dairy industry is to assure consumers of the safety of food products from dairy cattle (milk and dairy products and meat products).

Salmonella spp. transmission is a major issue affecting animal and public health; *Salmonella* spp. cause an estimated 696,000 to 3,840,000 human cases of foodborne disease and 696 to 3840 deaths annually in the US (9). As the pathogen monitored under the new HACCP (Hazard Analysis and Critical Control Point) regulations in slaughter plants, *Salmonella* spp. are starting to receive increased attention on the farm.

In addition, there are growing concerns for antimicrobial-resistant strains of salmonella. In the

United Kingdom, *Salmonella typhimurium* phage type DT104, with a characteristic pattern of resistance to multiple antibiotics (e.g., ampicillin, chloramphenicol, streptomycin, sulfonamides, and tetracyclines), has been associated with outbreaks of disease in humans and is now reported to be the second most prevalent salmonella in humans in that country (38). A case-control study showed that cases in humans were linked to the consumption of several foods as well as to contact with sick animals, including farm animals (50). This phage type has been identified in a limited number of human disease outbreaks in the US (17). Monitoring systems designed to evaluate changes in antimicrobial patterns in salmonella in US animal and human populations have been implemented (2).

Salmonella spp. have been identified from many livestock and poultry species. The NAHMS Dairy '96 study (53) showed that 27.5% of 91 US dairy operations from 19 states were shedding *Salmonella* spp. (56.5% of herds with at least 400 dairy cows). Although only 5% of dairy cows were culture-positive, the percentage of culture-positive dairy operations is an indication of the wide exposure of dairy cows to *Salmonella* spp.; fecal shedding is similar to that of US cattle on feed (40). Concepts related to the control of *Salmonella* spp. on dairy operations have been proposed (32) and need to be developed into best management practices for on-farm implementation.

Campylobacter jejuni is the most commonly isolated bacterial enteric pathogen of humans in the US, exceeding *Salmonella* spp. An estimated 2.1 million cases occur annually (34). The typical manifestations of enteritis are similar to those of other enteric pathogens, including diarrhea and cramping preceded by fever, headache, muscle pain, and malaise. The clinical signs are usually self-limiting, and recovery usually occurs within a week. Although sporadic cases of campylobacteriosis in humans are common, outbreaks rarely occur, perhaps because of widespread immunity in the human population resulting from repeated exposure (11). Vehicles of transmission to humans include untreated water and undercooked meat, including poultry meat, and raw milk. Contact with diarrheic pets is another method of transmission. Cross-contamination of food products also occurs, but transmission between humans seems uncommon.

The recently recognized association between development of Guillain-Barre syndrome in humans and prior *C. jejuni* infection, along with other sequelae, have increased the level of public health concern for this pathogen. Guillain-Barre syndrome, with a

reported annual incidence in the US of about 2 in 100,000 persons (34), is characterized by acute onset of paralysis, pain, and wasting muscles. Most Guillain-Barre patients recover completely with treatment, but some have severe, long-lasting complications (7). A recent case-control study (28) showed 26% of patients with Guillain-Barre syndrome had experienced a recent *C. jejuni* infection compared with 2% of control households. Other sequelae linked to *C. jejuni* include reactive arthritis and Reiter's syndrome. Although these sequelae have also been linked to a variety of other bacteria, viruses, vaccines, and drugs, *C. jejuni* is considered by some experts to be the chief cause of Guillain-Barre syndrome (7).

In livestock, the prevalence of *C. jejuni* has not been well defined. One study (29) of primarily Grade B dairies in Tennessee found that 12% of 292 samples from bulk milk tanks on the farm were culture-positive for *C. jejuni*. Another study (4) of 13 experimental farms in the Netherlands showed that 22% of samples of cow feces were culture-positive, another indication of the expected high level of *C. jejuni* on dairy operations. The risk factors for shedding *C. jejuni* have not been well evaluated to date, and specific preventive strategies have not been developed.

Cryptosporidium parvum represents another public concern of importance to the cattle industry because this pathogen survives chlorination of public water supplies and causes particular disease problems in immunocompromised humans (35). This organism is primarily shed in the feces of young animals, including cattle younger than 30 d of age. This age-associated shedding may be key to the development of effective preventive management.

In addition, considerable concern has been expressed related to the potential public health risk from other pathogens, including *M. paratuberculosis* and the agent causing bovine spongiform encephalopathy (5, 10). A great deal of uncertainty about these associations will continue until further scientific information becomes available.

Health Conditions and International Trade Concerns

Although accounting for only approximately 4% of the value of US milk produced, dairy-related exports totaled an estimated \$826 million in 1996 (49). Dairy product exports such as dry milk powder and cheese provided an overwhelming portion of this trade with a value of \$727 million; exports of live dairy cattle were valued at \$26 million. Trade in dairy cattle embryos

contributed an additional \$10 million to US exports, and exports of semen from all types of cattle added \$63 million. Defending these markets makes it important for the US dairy industry to address animal health issues that are a concern to the international community.

Creating opportunities for expanded US dairy-related exports reinforces this need. Because foreign markets are expected to increase their demand for dairy products at a greater rate than the domestic market, increased exports are viewed as a potential outlet for US production. Strong competition in these markets from exporters such as New Zealand will require the US to not only meet other countries' concerns regarding animal health but also create further incentives to lower production costs related to livestock diseases, agents, and conditions. Therefore, it is important for the US dairy industry to address animal health issues that are a concern to the international community.

For the purposes of international reporting, the Office International des Epizooties (**OIE**), Paris, France, with representation from each member country, has designated lists of transmissible diseases based on the potential for spread and economic or public health consequence. List A diseases have the potential for very serious and rapid spread and for serious socioeconomic or public health consequences; these diseases include diseases that are foreign to North America such as foot and mouth disease and rinderpest. One list A disease found in cattle in some parts of the US is bluetongue. List B diseases are considered to have socioeconomic or public health consequences within countries, but are less serious in nature than list A diseases. Examples of list B diseases include bovine tuberculosis, Johne's disease, and bovine leukosis virus. This classification system is useful to assess the importance of individual diseases for international trade, but the perceived importance of specific diseases varies by country.

Mycobacterium bovis, the agent causing bovine tuberculosis, has been nearly eradicated in US cattle, except for a very few dairy herds near the Mexican border (47). Efforts are ongoing to understand better transmission patterns as well as the potential reservoir species related to *M. bovis*. One possible reservoir of concern is wild cervids. A recent risk assessment (43) evaluating the risk of transmission from infected white-tailed deer in Michigan estimated a very low but serious risk to domestic cattle and discussed management options for dealing with this problem.

Although bluetongue typically causes only subclinical disease in cattle, bluetongue can cause severe clinical disease in sheep, goats, and wild ruminants.

Cattle are considered a major reservoir of the virus causing this disease. Bluetongue is an OIE list A disease and creates an impediment to trade of cattle with certain countries free of the virus, including those in the European Community. A national study has shown a high prevalence of antibody to bovine bluetongue virus in the southwestern US (23), and more recent annual surveys have consistently shown the northeastern and Great Lakes states to be low risk areas (46). This information, along with expanding knowledge of the life cycle of the *Culicoides* vector (37), has allowed movement of test-negative cattle from low risk areas during low risk seasons.

Another OIE list A disease of concern to US cattle operations is vesicular stomatitis. This disease is primarily important to the international trade community because of its similar clinical appearance to foot-and-mouth disease, though economic losses to affected dairy operations can be considerable, especially those caused by the premature culling of cows and by decreased milk production (1). Transmission of vesicular stomatitis virus is not completely understood, although transmission is thought to involve both direct contact and insect vectors. Cases in the US periodically appear in the warmer summer months. In 1995, positive premises were reported in 6 western states, resulting in some restrictions of international trade of US products (8).

The high prevalence of bovine leukosis virus (BLV), an OIE list B disease, on US dairy operations

also creates barriers to international trade of the infected cattle. The herd-level prevalence of BLV infection has been estimated to be 89% of US dairy herds (44). This prevalence is considered to be similar to that occurring in many other parts of the world (18) although higher than that in the European Community, where active control efforts have been underway for many years. Preventive strategies, including the reduction of potential blood transmission between animals, have been developed (14, 15), but the high prevalence of BLV virus in dairy cattle is an indication of the relative lack of adoption of these practices.

Health Conditions with Animal Welfare Concerns

Welfare concerns are growing in consumer recognition but are difficult to assess quantitatively. Rollin (30) has proposed that consumers of animal products have recognized changes in the nature of use of animals for agricultural purposes that have occurred with increased herd sizes and technologies of scale, and this recognition has led to a new ethic for the treatment of animals. This ethic is concerned with the welfare of all animals that suffer, not just those that suffer as a result of cruelty.

Particular concern seems to be directed toward diseases leading to perceived suffering by animals or

TABLE 1. Key dairy health conditions prioritized by production loss, zoonotic risk, international trade, and animal welfare concerns.¹

Disease or pathogen	Production loss	Zoonotic potential	International trade	Animal welfare
Mastitis	+++		+	
<i>Salmonella</i> spp.	+	++	+	
Johne's disease	++	ND ²	+ (B) ³	+
Bovine viral diarrhea	++		+	+
Reproductive diseases	++			
Lameness	++			+
Metabolic disease	++			+
<i>Cryptosporidium parvum</i>	+	++		
<i>Campylobacter jejuni</i>		++		
<i>Mycobacterium bovis</i>		+	+ (B)	
Bovine leukosis virus	+		+ (B)	
<i>Brucella abortus</i>		+	+ (B)	
BSE ²		ND	+ (B)	+
Bluetongue			+ (A)	
Vesicular stomatitis	+		+ (A)	

¹Relative ranking; +++ is more important than ++ or +.

²Not determined.

³The Office International des Epizooties has designated lists of transmissible diseases. List A diseases have the potential for very serious and rapid spread and for serious socioeconomic or public health consequences. Consequences of list B diseases are less serious.

⁴Bovine spongiform encephalopathy.

recumbency. Disease conditions of concern to the dairy industry include lameness and metabolic diseases such as milk fever, Johne's disease, and BVD. Clinical lameness in dairy cows has been recognized as an animal welfare concern because of the sometimes obvious signs of pain in the affected cattle (51). Periodic media reports describing cattle affected with diseases such as Johne's disease and BVD that may lead to weight loss and eventual emaciation or of the handling of recumbent cattle are also of public concern. Problems involved in handling recumbent cattle on the farm, whether caused by milk fever or other reasons, have led to the development of animal care guidelines for moving (when necessary) or euthanizing these cattle in humane ways, as part of the Milk and Dairy Beef Quality Assurance Program (22).

Prioritizing Dairy Health Conditions

Using the ranking method described, our assessment combining the four areas of productivity, zoonotic risk, international trade, and animal welfare ranked mastitis, *Salmonella*, Johne's disease, and BVD at the top of the list (Table 1). This evaluation illustrates the importance of markets to the dairy industry. If production-related losses alone are considered, key dairy health issues include mastitis, reproductive diseases, lameness, and metabolic diseases, as well as Johne's disease and BVD.

Prioritizing key dairy health concerns across production loss, zoonotic risk, international trade, and animal welfare categories should be helpful in assisting researchers and others in reevaluating criteria used in setting priorities. Other considerations, such as growing environmental concerns, were not part of this evaluation because they did not fit well within the animal health categorization format that we used for this evaluation. As the priorities for evaluation continue to shift through time, we strongly recognize the need to revisit this process, especially as more precise information becomes available.

After many years of work to prevent health conditions from causing production losses, the US has developed an abundant, inexpensive, and diverse food supply. In addition, the recognition of potential zoonotic risks has historically led to the development of a quality milk program (including dairy herd inspections and pasteurization as critical control steps). More recently, general concerns about food safety are expanding the list of diseases with zoonotic potential. For continuing expansion of the dairy industry, international trade will become more important. In order to compete in the world marketplace, the dairy indus-

try needs to meet the challenge of convincing consumers across the globe of the high level of quality and safety of US dairy products. Consumers are not only concerned about cost, diversity, quality, and safety, but also about production practices, including animal welfare issues.

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